# Data Analytics (DA)

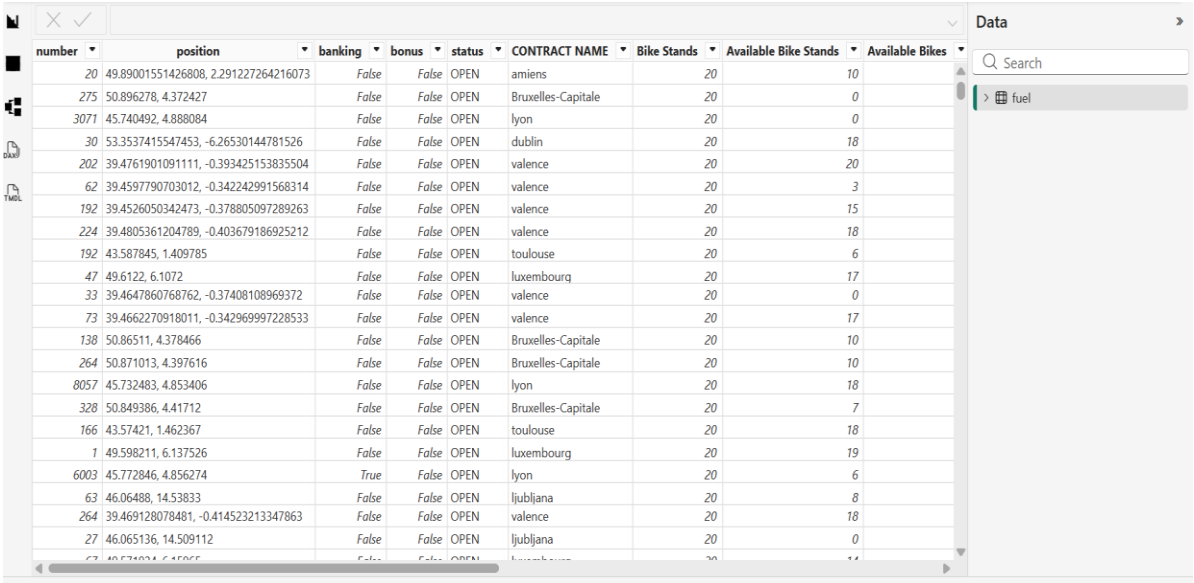
# Module-End -2

# Module End Assignment:

# “Data Transformation, Modeling & Visualization with Power BI”

## *TITLE: ANALYZING SMART CITY BIKE SHARING DATA USING POWER BI*

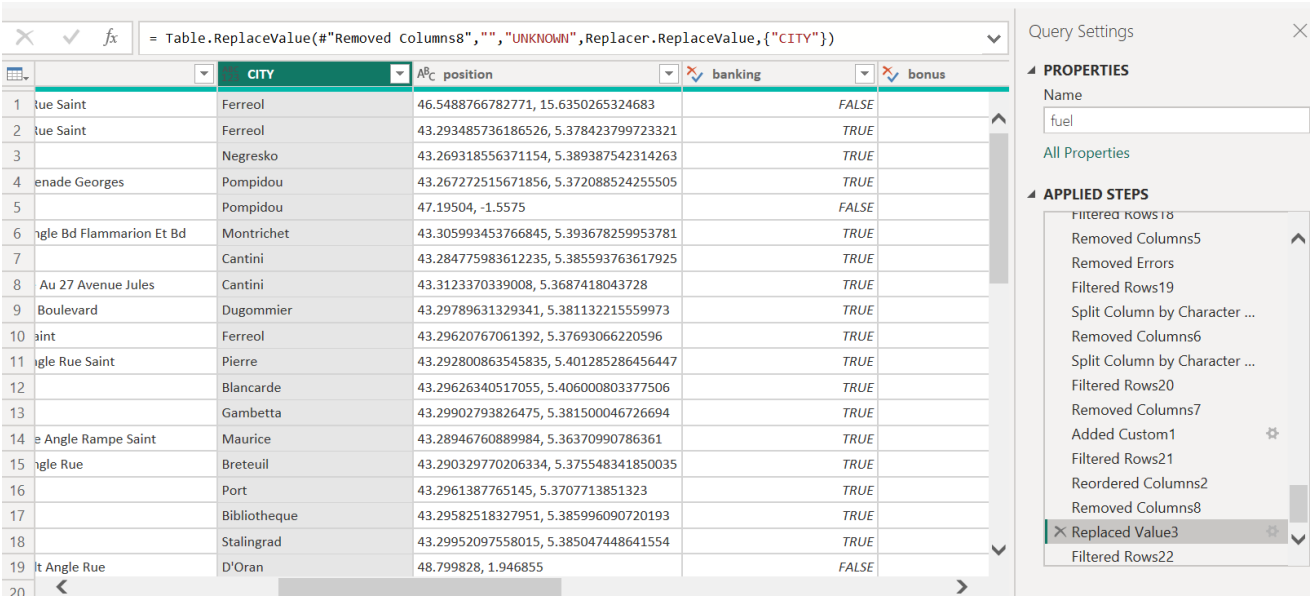
### DATA CONNECTION, IMPORT, INTERFACE EXPLORATION



| number | position | banking | bonus | status | CONTRACT NAME | Bike Stands | Available Bike Stands | Available Bikes |
|---|---|---|---|---|---|---|---|---|
| 20 | 49.89001551426808, 2.291227264216073 | False | False | OPEN | amiens | 20 | 10 | |
| 275 | 50.896278, 4.372427 | False | False | OPEN | Bruxelles-Capitale | 20 | 0 | |
| 3071 | 45.740492, 4.888084 | False | False | OPEN | lyon | 20 | 0 | |
| 30 | 53.3537415547453, -6.26530144781526 | False | False | OPEN | dublin | 20 | 18 | |
| 202 | 39.4761901091111, -0.393425153835504 | False | False | OPEN | valence | 20 | 20 | |
| 62 | 39.4597790703012, -0.342242991568314 | False | False | OPEN | valence | 20 | 3 | |
| 192 | 39.4526050342473, -0.378805097289263 | False | False | OPEN | valence | 20 | 15 | |
| 224 | 39.4805361204789, -0.403679186925212 | False | False | OPEN | valence | 20 | 18 | |
| 192 | 43.587845, 1.409785 | False | False | OPEN | toulouse | 20 | 6 | |
| 47 | 49.6122, 6.1072 | False | False | OPEN | luxembourg | 20 | 17 | |
| 33 | 39.4647860768762, -0.37408108969372 | False | False | OPEN | valence | 20 | 0 | |
| 73 | 39.4662270918011, -0.342969997228533 | False | False | OPEN | valence | 20 | 17 | |
| 138 | 50.86511, 4.378466 | False | False | OPEN | Bruxelles-Capitale | 20 | 10 | |
| 264 | 50.871013, 4.397616 | False | False | OPEN | Bruxelles-Capitale | 20 | 10 | |
| 8057 | 45.732483, 4.853406 | False | False | OPEN | lyon | 20 | 18 | |
| 328 | 50.849386, 4.41712 | False | False | OPEN | Bruxelles-Capitale | 20 | 7 | |
| 166 | 43.57421, 1.462367 | False | False | OPEN | toulouse | 20 | 18 | |
| 1 | 49.598211, 6.137526 | False | False | OPEN | luxembourg | 20 | 19 | |
| 6003 | 45.772846, 4.856274 | True | False | OPEN | lyon | 20 | 6 | |
| 63 | 46.06488, 14.53833 | False | False | OPEN | ljubljana | 20 | 8 | |
| 264 | 39.469128078481, -0.414523213347863 | False | False | OPEN | valence | 20 | 18 | |
| 27 | 46.065136, 14.509112 | False | False | OPEN | ljubljana | 20 | 0 | |

# DATA CLEANING & TRANSFORMATION

## ❖ Handled Missing Data



= Table.ReplaceValue(#"Removed Columns8","","UNKNOWN",Replacer.ReplaceValue,{"CITY"})

| | | CITY | position | banking | bonus |
|---|---|---|---|---|---|
| 1 | tue Saint | Ferreol | 46.5488766782771, 15.6350265324683 | FALSE | |
| 2 | tue Saint | Ferreol | 43.293485736186526, 5.378423799723321 | TRUE | |
| 3 | | Negresko | 43.269318556371154, 5.389387542314263 | TRUE | |
| 4 | enade Georges | Pompidou | 43.267272515671856, 5.372088524255505 | TRUE | |
| 5 | | Pompidou | 47.19504, -1.5575 | FALSE | |
| 6 | ngle Bd Flammarion Et Bd | Montrichet | 43.305993453766845, 5.393678259953781 | TRUE | |
| 7 | | Cantini | 43.284775983612235, 5.385593763617925 | TRUE | |
| 8 | Au 27 Avenue Jules | Cantini | 43.3123370339008, 5.3687418043728 | TRUE | |
| 9 | Boulevard | Dugommier | 43.29789631329341, 5.381132215559973 | TRUE | |
| 10 | aint | Ferreol | 43.29620767061392, 5.37693066220596 | TRUE | |
| 11 | ngle Rue Saint | Pierre | 43.292800863545835, 5.401285286456447 | TRUE | |
| 12 | | Blancarde | 43.29626340517055, 5.406000803377506 | TRUE | |
| 13 | | Gambetta | 43.29902793826475, 5.381500046726694 | TRUE | |
| 14 | e Angle Rampe Saint | Maurice | 43.28946760889984, 5.36370990786361 | TRUE | |
| 15 | ngle Rue | Breteuil | 43.290329770206334, 5.375548341850035 | TRUE | |
| 16 | | Port | 43.2961387765145, 5.3707713851323 | TRUE | |
| 17 | | Bibliotheque | 43.29582518327951, 5.385996090720193 | TRUE | |
| 18 | | Stalingrad | 43.29952097558015, 5.385047448641554 | TRUE | |
| 19 | t Angle Rue | D'Oran | 48.799828, 1.946855 | FALSE | |

Query Settings

PROPERTIES

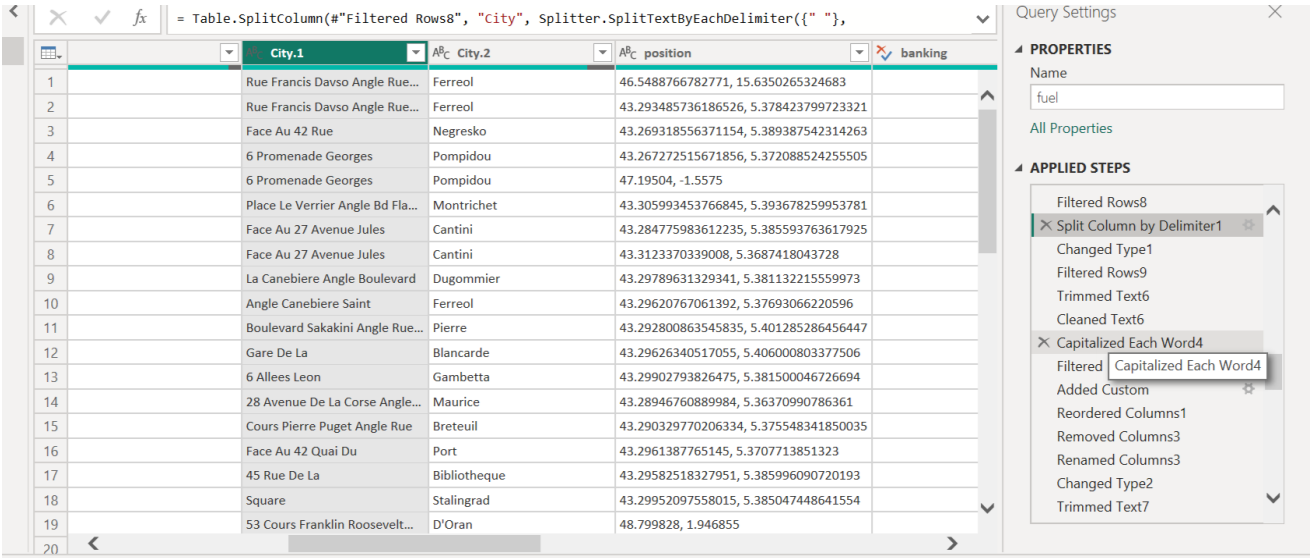
Name: fuel

All Properties

APPLIED STEPS

- Filtered Rows18
- Removed Columns5
- Removed Errors
- Filtered Rows19
- Split Column by Character ...
- Removed Columns6
- Split Column by Character ...
- Filtered Rows20
- Removed Columns7
- Added Custom1
- Filtered Rows21
- Reordered Columns2
- Removed Columns8
- Replaced Value3
- Filtered Rows22

## ❖ Formatted & Transformed Columns in Power Query



= Table.SplitColumn(#"Filtered Rows8", "City", Splitter.SplitTextByEachDelimiter({" "},

| | | City.1 | City.2 | position | banking |
|---|---|---|---|---|---|
| 1 | | Rue Francis Davso Angle Rue... | Ferreol | 46.5488766782771, 15.6350265324683 | |
| 2 | | Rue Francis Davso Angle Rue... | Ferreol | 43.293485736186526, 5.378423799723321 | |
| 3 | | Face Au 42 Rue | Negresko | 43.269318556371154, 5.389387542314263 | |
| 4 | | 6 Promenade Georges | Pompidou | 43.267272515671856, 5.372088524255505 | |
| 5 | | 6 Promenade Georges | Pompidou | 47.19504, -1.5575 | |
| 6 | | Place Le Verrier Angle Bd Fla... | Montrichet | 43.305993453766845, 5.393678259953781 | |
| 7 | | Face Au 27 Avenue Jules | Cantini | 43.284775983612235, 5.385593763617925 | |
| 8 | | Face Au 27 Avenue Jules | Cantini | 43.3123370339008, 5.3687418043728 | |
| 9 | | La Canebiere Angle Boulevard | Dugommier | 43.29789631329341, 5.381132215559973 | |
| 10 | | Angle Canebiere Saint | Ferreol | 43.29620767061392, 5.37693066220596 | |
| 11 | | Boulevard Sakakini Angle Rue... | Pierre | 43.292800863545835, 5.401285286456447 | |
| 12 | | Gare De La | Blancarde | 43.29626340517055, 5.406000803377506 | |
| 13 | | 6 Allees Leon | Gambetta | 43.29902793826475, 5.381500046726694 | |
| 14 | | 28 Avenue De La Corse Angle... | Maurice | 43.28946760889984, 5.36370990786361 | |
| 15 | | Cours Pierre Puget Angle Rue | Breteuil | 43.290329770206334, 5.375548341850035 | |
| 16 | | Face Au 42 Quai Du | Port | 43.2961387765145, 5.3707713851323 | |
| 17 | | 45 Rue De La | Bibliotheque | 43.29582518327951, 5.385996090720193 | |
| 18 | | Square | Stalingrad | 43.29952097558015, 5.385047448641554 | |
| 19 | | 53 Cours Franklin Roosevelt... | D'Oran | 48.799828, 1.946855 | |

Query Settings

PROPERTIES

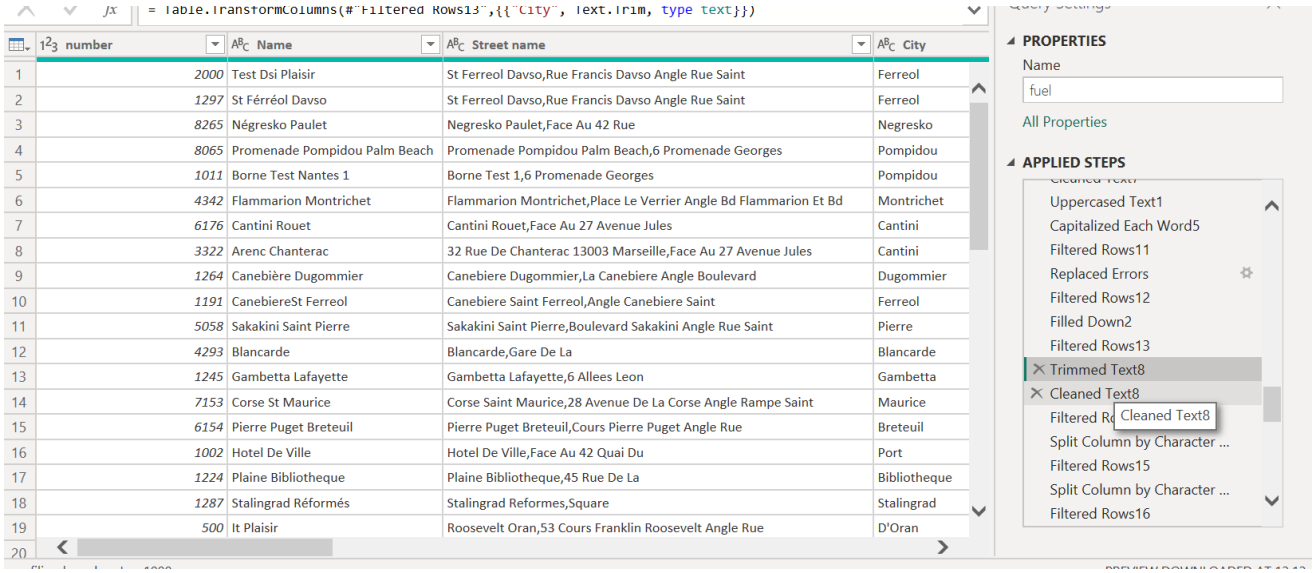
Name: fuel

All Properties

APPLIED STEPS

- Filtered Rows8
- Split Column by Delimiter1
- Changed Type1
- Filtered Rows9
- Trimmed Text6
- Cleaned Text6
- Capitalized Each Word4
- Filtered (Capitalized Each Word4)
- Added Custom
- Reordered Columns1
- Removed Columns3
- Renamed Columns3
- Changed Type2
- Trimmed Text7

❖ Trimmed, cleaned values by transform option

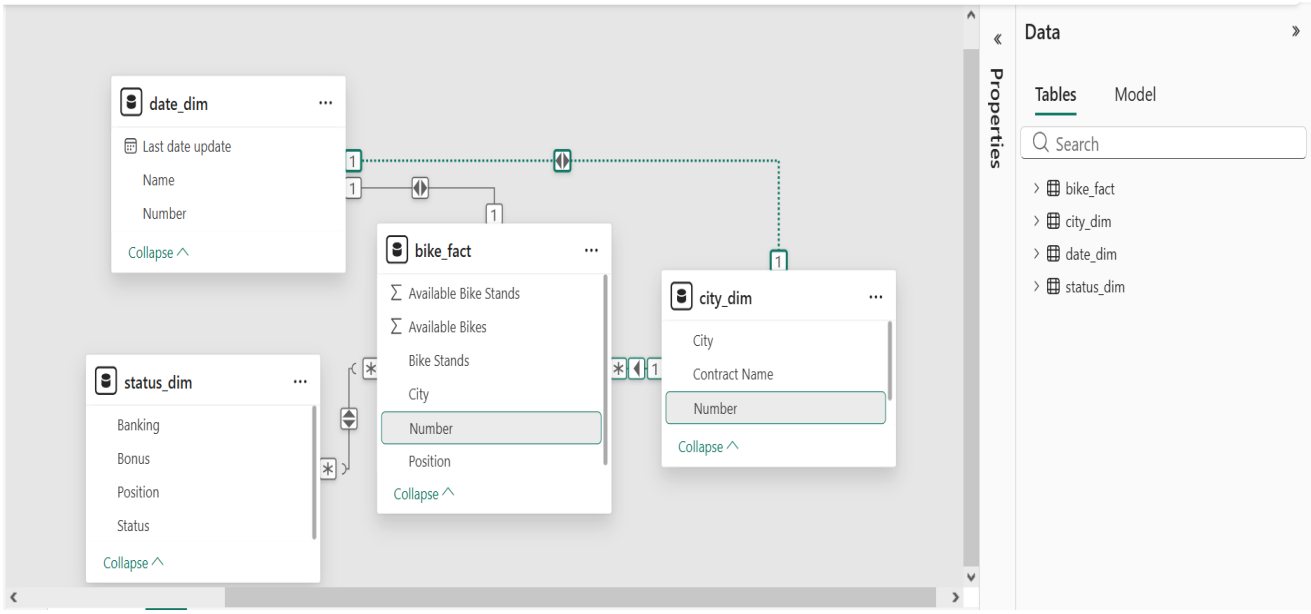


| | number | Name | Street name | City |
|---|---|---|---|---|
| 1 | 2000 | Test Dsi Plaisir | St Ferreol Davso,Rue Francis Davso Angle Rue Saint | Ferreol |
| 2 | 1297 | St Férréol Davso | St Ferreol Davso,Rue Francis Davso Angle Rue Saint | Ferreol |
| 3 | 8265 | Négresko Paulet | Negresko Paulet,Face Au 42 Rue | Negresko |
| 4 | 8065 | Promenade Pompidou Palm Beach | Promenade Pompidou Palm Beach,6 Promenade Georges | Pompidou |
| 5 | 1011 | Borne Test Nantes 1 | Borne Test 1,6 Promenade Georges | Pompidou |
| 6 | 4342 | Flammarion Montrichet | Flammarion Montrichet,Place Le Verrier Angle Bd Flammarion Et Bd | Montrichet |
| 7 | 6176 | Cantini Rouet | Cantini Rouet,Face Au 27 Avenue Jules | Cantini |
| 8 | 3322 | Arenc Chanterac | 32 Rue De Chanterac 13003 Marseille,Face Au 27 Avenue Jules | Cantini |
| 9 | 1264 | Canebière Dugommier | Canebiere Dugommier,La Canebiere Angle Boulevard | Dugommier |
| 10 | 1191 | CanebiereSt Ferreol | Canebiere Saint Ferreol,Angle Canebiere Saint | Ferreol |
| 11 | 5058 | Sakakini Saint Pierre | Sakakini Saint Pierre,Boulevard Sakakini Angle Rue Saint | Pierre |
| 12 | 4293 | Blancarde | Blancarde,Gare De La | Blancarde |
| 13 | 1245 | Gambetta Lafayette | Gambetta Lafayette,6 Allees Leon | Gambetta |
| 14 | 7153 | Corse St Maurice | Corse Saint Maurice,28 Avenue De La Corse Angle Rampe Saint | Maurice |
| 15 | 6154 | Pierre Puget Breteuil | Pierre Puget Breteuil,Cours Pierre Puget Angle Rue | Breteuil |
| 16 | 1002 | Hotel De Ville | Hotel De Ville,Face Au 42 Quai Du | Port |
| 17 | 1224 | Plaine Bibliotheque | Plaine Bibliotheque,45 Rue De La | Bibliotheque |
| 18 | 1287 | Stalingrad Réformés | Stalingrad Reformes,Square | Stalingrad |
| 19 | 500 | It Plaisir | Roosevelt Oran,53 Cours Franklin Roosevelt Angle Rue | D'Oran |



## Data Modeling

- Built Fact & Dimension Tables, Defined Relationships, Create Star Schema



## DAX & Calculations

- ✓ **Created calculated measures:**
  - TOTAL_BIKESTANDS = SUM (BIKE_ FACT [BIKE STANDS])
  - TOTAL AVL_BIKES = SUM (BIKE_ FACT [AVAILABLE BIKES])
  - AVL_STANDS = [TOTAL_BIKESTANDS]- [TOTAL AVL_BIKES]
  - OCCUPANCY % = (DIVIDE ([TOTAL AVL_BIKES],[TOTAL_BIKESTANDS],0))
  - DAYS SINCE_UPDATE = DATEDIFF (MAX (DATE_DIM [DATE UPDATED]), TODAY (), DAY)

## Descriptive Statistics

- ✓ The average (13.86) is significantly higher than the mean/median (8.00).
- ✓ The **average (13.86)** is significantly higher than the **mean/median (8.00)**.
- ✓ The **Standard Deviation (7.37)** for available bikes is high relative to the mean. This confirms that bike distribution is inconsistent across the network, with some stations potentially full and others completely empty.

| TOTAL_BIKESTANDS |
|---|
| 146 |

| OCCUPANCY % |
|---|
| 30.14% |

| TOTAL AVL_BIKES |
|---|
| 44 |

| AVL_STANDS |
|---|
| 102 |

| DAYS SINCE_UPDATE |
|---|
| |

| avg_bikestands |
|---|
| 13.86 |

| mean_bikestands |
|---|
| 8.00 |

| std_avl bikes |
|---|
| 7.37 |

✓ **Created calculated columns:**

**"Remaining bike stands"** column created

```
REMAINING BIKE STANDS = bike_fact[Bike Stands]-bike_fact[Available Bike Stands]
```

| Available Bike Stands | Available Bikes | Position | Number | City | Last date update | REMAINING BIKE STANDS |
|---|---|---|---|---|---|---|
| 0 | 0 | 50.867643, 4.385817 | 164 | UNKNOWN | 20-01-2025 05:34:14 | 25 |
| 10 | 0 | 49.89001551426808, 2.291227264216073 | 20 | UNKNOWN | 11-06-2025 12:34:42 | 10 |
| 11 | 0 | 47.744426, 7.340096 | 18 | UNKNOWN | 02-04-2025 12:31:59 | 0 |
| 0 | 0 | 50.896278, 4.372427 | 275 | UNKNOWN | 07-10-2025 10:53:43 | 20 |
| 0 | 0 | 50.840942, 4.436529 | 341 | UNKNOWN | 30-07-2024 12:51:29 | 23 |
| 0 | 0 | 50.863388, 4.386222 | 157 | UNKNOWN | 01-08-2025 10:57:29 | 24 |
| 6 | 6 | 45.772277, 4.865774 | 10075 | UNKNOWN | 15-09-2025 14:38:08 | 6 |
| 0 | 0 | 45.753528, 4.857784 | 3040 | UNKNOWN | 27-02-2024 13:37:16 | 35 |
| 0 | 0 | 50.847649, 4.307714 | 198 | UNKNOWN | 24-01-2024 11:00:08 | 25 |
| 0 | 1 | 50.4773882, 4.8419898 | 1001 | UNKNOWN | 05-04-2024 12:14:23 | 1 |
| 0 | 0 | 45.749257, 4.82984 | 2039 | UNKNOWN | 24-01-2024 16:37:25 | 17 |
| 1 | 0 | 48.7998343395716, 1.94656723572685 | 511 | UNKNOWN | 23-09-2024 19:52:42 | 0 |
| 17 | 8 | 50.809799, 4.317546 | 234 | UNKNOWN | 22-11-2024 18:10:27 | 8 |
| 0 | 0 | 47.74409332327696, 7.333486550613367 | 19 | UNKNOWN | 06-02-2025 18:43:17 | 14 |
| 0 | 0 | 50.893497, 4.363297 | 304 | UNKNOWN | 28-04-2025 10:46:31 | 25 |
| 0 | 1 | 49.894393, 2.266713 | 50 | UNKNOWN | 05-05-2025 14:28:04 | 1 |
| 19 | 5 | 50.869297, 4.36257 | 159 | UNKNOWN | 12-11-2025 14:47:35 | 6 |
| 0 | 0 | 45.740492, 4.888084 | 3071 | UNKNOWN | 12-11-2025 14:45:10 | 20 |
| 18 | 2 | 53.3537415547453, -6.26530144781526 | 30 | UNKNOWN | 12-11-2025 14:47:57 | 2 |

Data

Search

- bike_fact
  - Available Bike Stands
  - Available Bikes
  - Bike Stands
  - City
  - Last date update
  - Number
  - Position
  - REMAINING BIKE STANDS
- city_dim
  - City
  - Contract Name
  - Number
  - Street Name
- date_dim
  - DATE UPDATED

"**Bike status**" column created

```
BIKE_STATUS = IF(bike_fact[Available Bikes]<=0,"EMPTY",
IF(bike_fact[Available Bikes]<10,"LOW","HIGH"))
```

| | Available Bikes | Position | Number | City | Last date update | REMAINING BIKE STANDS | BIKE_STATUS |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 50.867643, 4.385817 | 164 | UNKNOWN | 20-01-2025 05:34:14 | 25 | EMPTY |
| 0 | 0 | 49.89001551426808, 2.291227264216073 | 20 | UNKNOWN | 11-06-2025 12:34:42 | 10 | EMPTY |
| 1 | 0 | 47.744426, 7.340096 | 18 | UNKNOWN | 02-04-2025 12:31:59 | 0 | EMPTY |
| 0 | 0 | 50.896278, 4.372427 | 275 | UNKNOWN | 07-10-2025 10:53:43 | 20 | EMPTY |
| 0 | 0 | 50.840942, 4.436529 | 341 | UNKNOWN | 30-07-2024 12:51:29 | 23 | EMPTY |
| 0 | 0 | 50.863388, 4.386222 | 157 | UNKNOWN | 01-08-2025 10:57:29 | 24 | EMPTY |
| 6 | 6 | 45.772277, 4.865774 | 10075 | UNKNOWN | 15-09-2025 14:38:08 | 6 | LOW |
| 0 | 0 | 45.753528, 4.857784 | 3040 | UNKNOWN | 27-02-2024 13:37:16 | 35 | EMPTY |
| 0 | 0 | 50.847649, 4.307714 | 198 | UNKNOWN | 24-01-2024 11:00:08 | 25 | EMPTY |
| 0 | 1 | 50.4773882, 4.8419898 | 1001 | UNKNOWN | 05-04-2024 12:14:23 | 1 | LOW |
| 0 | 0 | 45.749257, 4.82984 | 2039 | UNKNOWN | 24-01-2024 16:37:25 | 17 | EMPTY |
| 1 | 0 | 48.7998343395716, 1.94656723572685 | 511 | UNKNOWN | 23-09-2024 19:52:42 | 0 | EMPTY |
| 7 | 8 | 50.809799, 4.317546 | 234 | UNKNOWN | 22-11-2024 18:10:27 | 8 | LOW |
| 0 | 0 | 47.74409332327696, 7.333486550613367 | 19 | UNKNOWN | 06-02-2025 18:43:17 | 14 | EMPTY |
| 0 | 0 | 50.893497, 4.363297 | 304 | UNKNOWN | 28-04-2025 10:46:31 | 25 | EMPTY |
| 0 | 1 | 49.894393, 2.266713 | 50 | UNKNOWN | 05-05-2025 14:28:04 | 1 | LOW |
| 9 | 5 | 50.869297, 4.36257 | 159 | UNKNOWN | 12-11-2025 14:47:35 | 6 | LOW |
| 0 | 0 | 45.740492, 4.888084 | 3071 | UNKNOWN | 12-11-2025 14:45:10 | 20 | EMPTY |
| 8 | 2 | 53.3537415547453, -6.26530144781526 | 30 | UNKNOWN | 12-11-2025 14:47:57 | 2 | LOW |

“**Date diffrence with years** “column created

```
datediff_years = datediff(date_dim[DATE UPDATED],today(),YEAR)
```

| Number | Name | DATE UPDATED | datediff_years |
|---|---|---|---|
| 262 | Avda De Dinamarca | 12-11-2025 14:48:56 | 1 |
| 297 | Tenreuken | 12-11-2025 14:49:00 | 1 |
| 117 | Place Du Cirque | 12-11-2025 14:49:13 | 1 |
| 254 | AvdaDrTomasSalaCarteros | 12-11-2025 14:49:11 | 1 |
| 263 | Maison Communale D'Evere or Gemeer | 12-11-2025 14:49:14 | 1 |
| 140 | AvenidaCampanar | 12-11-2025 14:48:55 | 1 |
| 42 | Ramiro Maetzu | 12-11-2025 14:49:23 | 1 |
| 150 | Meiser | 12-11-2025 14:49:13 | 1 |
| 10086 | Anatole France | 12-11-2025 14:49:17 | 1 |
| 151 | Curie Négreneys | 12-11-2025 14:49:05 | 1 |
| 171 | Piscine Louis NamecheorZwembad Lou | 12-11-2025 14:49:17 | 1 |
| 336 | Colonne Salonique | 12-11-2025 14:49:00 | 1 |
| 40 | Gare De Dornach | 26-03-2025 13:41:30 | 1 |
| 23 | Hasenrain | 26-03-2025 12:29:02 | 1 |
| 41 | Station Solaire | 25-03-2025 14:33:58 | 1 |
| 5060 | Sakakini Chave | 13-09-2022 11:15:06 | 4 |
| 10121 | Parc Du Centre | 04-09-2023 14:10:44 | 3 |
| 109 | Avenida San Francisco Javier | 02-10-2023 10:06:41 | 3 |
| 600 | Run In LyonSolaire | 18-01-2022 21:58:12 | 4 |
| 300 | Auberge | 17-08-2023 13:10:23 | 3 |

Data

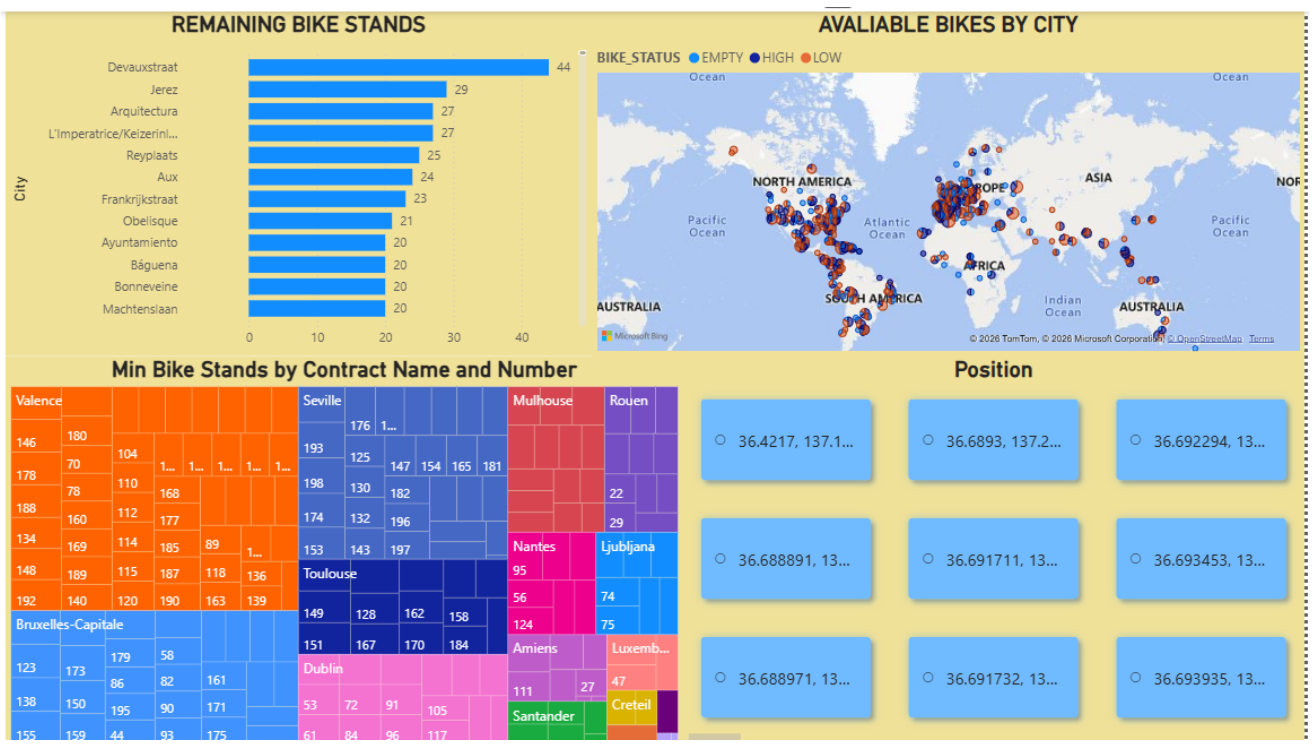
Search

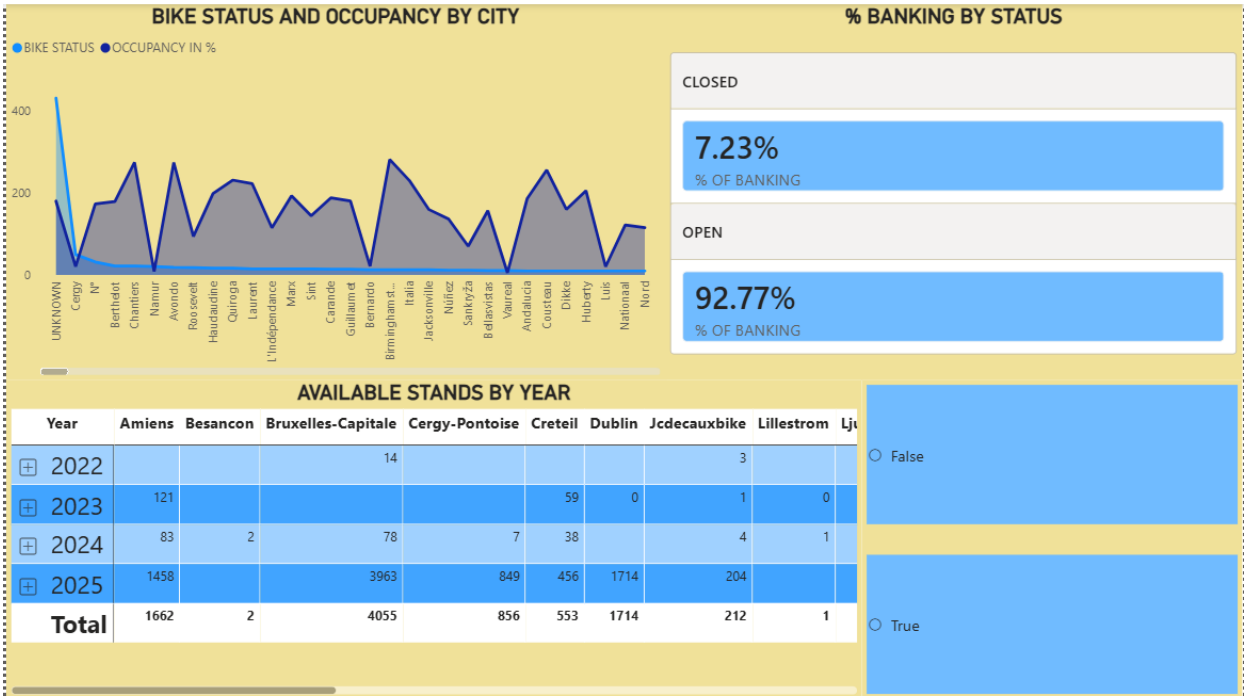
Contract Name
Number
Street Name
date_dim
DATE UPDATED
datediff_years
Name
Number
measure_table
AVL_STANDS
Column
DAYS SINCE_UPDATE
OCCUPANCY %
TOTAL AVL_BIKES
TOTAL_BIKESTANDS
status_dim
Banking

## DATA VISUALIZATION

### INSIGHTS:

- ✓ **Devauxstraat** is the highest-performing station for bike returns with **44 available stands**.
- ✓ Bike-sharing demand is heavily concentrated in **Europe and coastal North/South America**
- ✓ The Tree map highlights **Valence** and **Bruxelles-Capitale** as the largest contract networks,



## INSIGHTS:

- ✓ The data indicates a shift from a fragmented network in 2022–2024 to a **high-density, large-scale operation in 2025**, primarily led by the Brussels and Dublin contracts.
- ✓ While most cities peaked in 2025, regions like **Besancon** and **Lillestrom** remain in early pilot phases with fewer than 5 stands.
- ✓ **92.77%** of the banking system is **Open**, indicating a healthy, functional financial transaction layer for the bike-share system.